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TREND-BASED MOCK PAPER

GATE Textile Engineering and Fibre Science

Mock Paper — 04

Full-Length Examination Simulation

Duration: 3 Hours | Maximum Marks: 100 | Questions: 65

Examination Details

Branch:	Textile Engineering and Fibre Science
Paper:	Mock 04
Total Questions:	65
Total Marks:	100
Duration:	3 Hours
Question Types:	MCQ, MSQ, NAT
Negative Marking:	Yes (MCQ only)
Difficulty:	Mixed

Unofficial practice material. Not affiliated with or endorsed by GATE/IIT.
Historical patterns do not guarantee future examination questions.
This is a trend-based practice simulation.

GENERAL INSTRUCTIONS

1. This question paper contains 65 questions for a total of 100 marks.
2. Duration of examination: 3 Hours.
3. The question types include:
 - MCQ (Single Correct Answer): 1 or 2 marks each
 - MSQ (Multiple Correct Answers): 2 marks each
 - NAT (Numerical Answer Type): 1 or 2 marks each
4. For MCQ: Only one option is correct. Select the single correct answer.
5. For MSQ: One or more options are correct. Select ALL correct options.
6. For NAT: Enter the numerical answer. No negative marking for NAT questions.
7. Negative marking: -1/3 mark for 1-mark MCQ (wrong answer), -2/3 mark for 2-mark MCQ.
No negative marking for MSQ and NAT questions.
8. A virtual calculator is available on the screen. Physical calculators are not permitted.
9. The question paper contains 8 pages including this instructions page.
10. All questions are compulsory.
11. Fill in the required details on the answer sheet before starting.
12. Do not ask for clarifications from the invigilator. Interpret questions as given.

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Section A: General Aptitude

Q1. [1-MCQ] [1 mark]

Perimeter of square side 5:

- (A) 10
- (B) 20
- (C) 25
- (D) 5

Section B: Textile Engineering and Fibre Science

Q2. [1-MCQ] [1 mark]

Integral of $2x \, dx$:

- (A) $x^2 + C$
- (B) $2x + C$
- (C) $x + C$
- (D) $x^{2/2}$

Q3. [1-MCQ] [1 mark]

$\lim_{n \rightarrow \infty} (1 + 1/n)^{2n}$ equals:

- (A) e
- (B) e^2
- (C) 1
- (D) infinity

Q4. [1-MCQ] [1 mark]

Rolle's theorem requires $f(x)$ to be:

- (A) Continuous only
- (B) Differentiable only
- (C) Continuous on $[a,b]$ and differentiable on (a,b)

(D) Both continuous and differentiable everywhere

Q5. [1-MCQ] [1 mark]

$\frac{d}{dx}[\ln(\sec x) + \ln(\tan x)]$ evaluated at $x = \pi/4$ equals:

- (A) 0
- (B) 2
- (C) 4
- (D) 1

Q6. [1-MCQ] [1 mark]

Consider $f(x) = |x|$ on $[-1,1]$. This function is:

- (A) Differentiable everywhere
- (B) Continuous but not differentiable at 0
- (C) Neither continuous nor differentiable
- (D) Differentiable at 0 only

Q7. [1-NAT] [1 mark]

Fourier series of $f(x)=x$ on $(-\pi,\pi)$ has only:

(Numerical Answer Type — Enter the numerical value)

Q8. [1-MCQ] [1 mark]

The Laplace transform of $t \cdot e^{-2t}$ is:

- (A) $1/(s+2)^2$
- (B) $1/(s-2)^2$
- (C) $s/(s+2)^2$
- (D) $2/(s+2)^2$

Q9. [1-MCQ] [1 mark]

$P(A \cup B) = 0.6$, $P(A \cap B) = 0.2$, $P(B) = 0.4$. Then $P(A|B)$ equals:

- (A) 0.2
- (B) 0.5
- (C) 0.6
- (D) 0.4

Q10. [1-NAT] [1 mark]

Laplace transform of $e^{-3t} \cdot u(t)$ is:

(Numerical Answer Type — Enter the numerical value)

Q11. [1-MCQ] [1 mark]

If $f(x) = x^3 - 3x + 2$, number of real roots:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Q12. [1-NAT] [1 mark]

Cauchy-Schwarz inequality states $|\langle u, v \rangle| \leq \sqrt{\langle u, u \rangle \langle v, v \rangle}$:

(Numerical Answer Type — Enter the numerical value)

Q13. [1-NAT] [1 mark]

Gauss-Seidel method converges when the coefficient matrix is:

(Numerical Answer Type — Enter the numerical value)

Q14. [1-MCQ] [1 mark]

Number of spanning trees in complete graph K_4 :

- (A) 4
 - (B) 8
 - (C) 16
 - (D) 12
-

Q15. [1-MCQ] [1 mark]

$P(|Z| \leq 1)$ for $Z \sim N(0,1)$ is approximately:

- (A) 0.68
 - (B) 0.32
 - (C) 0.95
 - (D) 0.16
-

Q16. [1-NAT] [1 mark]

Solution of $y'' + 4y = 0$ with $y(0)=1$, $y'(0)=0$, $y(\pi/2)$ equals:

(Numerical Answer Type — Enter the numerical value)

Q17. [2-NAT] [2 marks]

If a 3×3 matrix has eigenvalues 1, 2, and 3, then $\text{trace}(A^2)$ equals:

(Numerical Answer Type — Enter the numerical value)

Q18. [2-MCQ] [2 marks]

The area of a circle of radius r is:

- (A) πr^2
 - (B) $2\pi r$
 - (C) πr
 - (D) $2\pi r^2$
-

Q19. [2-NAT] [2 marks]

A random variable $X \sim \text{Poisson}(\lambda=4)$. $P(X \geq 2)$ is approximately:

(Numerical Answer Type — Enter the numerical value)

Q20. [2-MSQ] [2 marks]

The rank of the matrix $\begin{bmatrix} 1, 2, 3 \\ 2, 4, 6 \\ 3, 6, 9 \end{bmatrix}$ is:

- (A) 1
 - (B) 2
 - (C) 3
 - (D) 0
-

Q21. [2-MSQ] [2 marks]

The differential equation $y \, dx - x \, dy = 0$ has solution:

- (A) $y = Cx$
- (B) $y = C/x$
- (C) $y = Cx^2$
- (D) $x^2 + y^2 = C$

Q22. [2-MCQ] [2 marks]

For matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $A^{50} \cdot [1, 0]^T$ equals:

- (A) $[1, 0]^T$
- (B) $[2, 1]^T$
- (C) $[(3+2\sqrt{2})^{50}, \dots]$
- (D) Cannot determine

Q23. [2-NAT] [2 marks]

Which of the following is a probability generating function for Poisson(λ)?

(Numerical Answer Type — Enter the numerical value)

Q24. [2-MCQ] [2 marks]

For a Poisson distribution with mean 4, variance is:

- (A) 4
- (B) 2
- (C) 16
- (D) 1

Q25. [2-MCQ] [2 marks]

If $f(z) = z^2 + 3z + 2$ is analytic, then $\operatorname{Re}(f(i)) =$

- (A) -1
- (B) -2
- (C) 1
- (D) 2

Q26. [2-NAT] [2 marks]

For the ODE $dy/dx = y/x + \sqrt{y^2/x^2}$, with $y(1)=0$, the solution is:

(Numerical Answer Type — Enter the numerical value)

Q27. [2-MCQ] [2 marks]

The vector $(1, 2, 3)$ projected onto $(2, 1, 0)$ is:

- (A) $(4/5, 2/5, 0)$
- (B) $(8/5, 4/5, 0)$
- (C) $(2, 1, 0)$
- (D) $(1, 2, 3)$

Q28. [2-MCQ] [2 marks]

For matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $A^{50} \cdot [1, 0]^T$ equals:

- (A) $[1, 0]^T$
- (B) $[2, 1]^T$
- (C) $[(3+2\sqrt{2})^{50}, \dots]$
- (D) Cannot determine

Q29. [2-MCQ] [2 marks]

The differential equation $y \, dx - x \, dy = 0$ has solution:

- (A) $y = C \cdot x$
- (B) $y = C/x$
- (C) $y = C \cdot x^2$
- (D) $x^2 + y^2 = C$

Q30. [2-NAT] [2 marks]

For a Poisson distribution with mean 4, variance is:

(Numerical Answer Type — Enter the numerical value)

Q31. [2-MCQ] [2 marks]

$P(|Z| \leq 1)$ for $Z \sim N(0,1)$ is approximately:

- (A) 0.68
- (B) 0.32
- (C) 0.95
- (D) 0.16

Q32. [2-MCQ] [2 marks]

If a 3×3 matrix has eigenvalues 1, 2, and 3, then $\text{trace}(A^2)$ equals:

- (A) 14
- (B) 6
- (C) 36
- (D) 12

Q33. [2-MCQ] [2 marks]

Laplace transform of $e^{-3t}u(t)$ is:

- (A) $1/(s+3)$
- (B) $1/(s-3)$
- (C) $1/s$
- (D) e^{-3}/s

Q34. [1-MCQ] [1 mark]

Spinning converts:

- (A) Fiber to yarn
- (B) Yarn to fabric
- (C) Fabric to garment
- (D) Fiber to fabric

Q35. [1-MCQ] [1 mark]

Weaving interlaces:

- (A) Warp and weft
- (B) Warp only
- (C) Weft only
- (D) Yarn only

Q36. [2-MCQ] [2 marks]

Spinning converts:

- (A) Fiber to yarn
- (B) Yarn to fabric
- (C) Fabric to garment
- (D) Fiber to fabric

Q37. [2-MSQ] [2 marks]

Weaving interlaces:

- (A) Warp and weft
- (B) Warp only
- (C) Weft only

(D) Yarn only

Q38. [2-MCQ] [2 marks]

Spinning converts:

- (A) Fiber to yarn
- (B) Yarn to fabric
- (C) Fabric to garment
- (D) Fiber to fabric

Q39. [2-MCQ] [2 marks]

Weaving interlaces:

- (A) Warp and weft
- (B) Warp only
- (C) Weft only
- (D) Yarn only

Q40. [2-MCQ] [2 marks]

Weaving interlaces:

- (A) Warp and weft
- (B) Warp only
- (C) Weft only
- (D) Yarn only

Q41. [2-MCQ] [2 marks]

Spinning converts:

- (A) Fiber to yarn
- (B) Yarn to fabric
- (C) Fabric to garment
- (D) Fiber to fabric

Q42. [2-MCQ] [2 marks]

Spinning converts:

- (A) Fiber to yarn
- (B) Yarn to fabric
- (C) Fabric to garment
- (D) Fiber to fabric

Q43. [2-MSQ] [2 marks]

Weaving interlaces:

- (A) Warp and weft
- (B) Warp only
- (C) Weft only
- (D) Yarn only

Q44. [2-MCQ] [2 marks]

Weaving interlaces:

- (A) Warp and weft
- (B) Warp only
- (C) Weft only
- (D) Yarn only

Q45. [2-NAT] [2 marks]

Spinning converts:

(Numerical Answer Type — Enter the numerical value)

Q46. [1-MCQ] [1 mark]

Cotton is a:

- (A) Protein
 - (B) Cellulosic
 - (C) Synthetic
 - (D) Mineral
-

Q47. [1-MCQ] [1 mark]

Moisture regain of cotton approx:

- (A) 8%
 - (B) 0.4%
 - (C) 12%
 - (D) 20%
-

Q48. [1-MCQ] [1 mark]

Cotton is a:

- (A) Animal fiber
 - (B) Vegetable fiber
 - (C) Mineral fiber
 - (D) Synthetic fiber
-

Q49. [1-MCQ] [1 mark]

Wool is classified as:

- (A) Vegetable
 - (B) Animal
 - (C) Mineral
 - (D) Synthetic
-

Q50. [1-NAT] [1 mark]

Nylon is a:

(Numerical Answer Type — Enter the numerical value)

Q51. [2-MCQ] [2 marks]

Moisture regain of cotton approx:

- (A) 8%
 - (B) 0.4%
 - (C) 12%
 - (D) 20%
-

Q52. [2-MCQ] [2 marks]

Wool is classified as:

- (A) Vegetable
 - (B) Animal
 - (C) Mineral
 - (D) Synthetic
-

Q53. [2-MCQ] [2 marks]

Moisture regain of cotton approx:

- (A) 8%
- (B) 0.4%
- (C) 12%
- (D) 20%

Q54. [2-MCQ] [2 marks]

Wool is classified as:

- (A) Vegetable
- (B) Animal
- (C) Mineral
- (D) Synthetic

Q55. [2-MCQ] [2 marks]

Moisture regain of cotton approx:

- (A) 8%
- (B) 0.4%
- (C) 12%
- (D) 20%

Q56. [2-MCQ] [2 marks]

Cotton is a:

- (A) Animal fiber
- (B) Vegetable fiber
- (C) Mineral fiber
- (D) Synthetic fiber

Q57. [2-NAT] [2 marks]

Wool is classified as:

(Numerical Answer Type — Enter the numerical value)

Q58. [2-MCQ] [2 marks]

Cotton is a:

- (A) Animal fiber
- (B) Vegetable fiber
- (C) Mineral fiber
- (D) Synthetic fiber

Q59. [1-NAT] [1 mark]

Carding does:

(Numerical Answer Type — Enter the numerical value)

Q60. [1-MCQ] [1 mark]

Plain weave has:

- (A) 1 up 1 down
- (B) 2 up 2 down
- (C) 3 up 1 down
- (D) Random

Q61. [1-MCQ] [1 mark]

Tensile strength measured by:

- (A) Microscope
- (B) Tensile tester
- (C) Air permeability
- (D) None

Q62. [1-MCQ] [1 mark]

Reactive dye bonds with:

- (A) Physical
- (B) Covalent
- (C) Hydrogen
- (D) Van der Waals

Q63. [1-MCQ] [1 mark]

Pattern is for:

- (A) Fabric design
- (B) Garment shape
- (C) Dyeing
- (D) Finishing

Q64. [1-MCQ] [1 mark]

Ring frame produces:

- (A) Sliver
- (B) Roving
- (C) Yarn
- (D) Fabric

Q65. [1-MCQ] [1 mark]

Coulomb friction law: $F = ?$

- (A) μN
 - (B) μF
 - (C) N/μ
 - (D) $\mu' N$
-

ANSWER KEY

Q. No.	Type	Answer	Marks
1	1MCQ	B	1
2	1MCQ	A	1
3	1MCQ	B	1
4	1NAT	A	1
5	1MCQ	A	1
6	1MCQ	B	1
7	1MCQ	B	1
8	1MCQ	B	1
9	1MCQ	C	1
10	1MCQ	A	1
11	1MCQ	B	1
12	1MCQ	C	1
13	1MCQ	A	1

14	1MCQ	C	1
15	1MCQ	A	1
16	1MCQ	A	1
17	1MCQ	B	1
18	1MCQ	B	1
19	1NAT	A	1
20	1MCQ	B	1
21	1NAT	B	1
22	1MCQ	A	1
23	1MCQ	B	1
24	1NAT	A	1
25	1MCQ	B	1
26	1NAT	A	1
27	1NAT	C	1
28	1MCQ	C	1

29	1MCQ	A	1
30	1NAT	C	1
31	2NAT	A	2
32	2MCQ	A	2
33	2NAT	A	2
34	2MSQ	A	2
35	2MSQ	B	2
36	2MCQ	A	2
37	2NAT	A	2
38	2MCQ	A	2
39	2MCQ	A	2
40	2NAT	A	2
41	2MCQ	B	2
42	2MCQ	A	2
43	2MCQ	A	2

44	2MCQ	B	2
45	2MSQ	A	2
46	2MCQ	A	2
47	2MCQ	A	2
48	2MCQ	A	2
49	2MCQ	B	2
50	2MCQ	A	2
51	2MCQ	B	2
52	2MCQ	A	2
53	2MCQ	A	2
54	2NAT	A	2
55	2MCQ	A	2
56	2MCQ	A	2
57	2MCQ	A	2
58	2MCQ	B	2

59	2MSQ	A	2
60	2MCQ	A	2
61	2MCQ	A	2
62	2MCQ	A	2
63	2NAT	A	2
64	2NAT	B	2
65	2MCQ	B	2

DETAILED SOLUTIONS

Q1.

Answer: B

Solution:

Correct Answer: B

Subject: General Aptitude | Topic: Quantitative | Difficulty: moderate

Q2.

Answer: A

Solution:

Correct Answer: A

Subject: Engineering Mathematics | Topic: Calculus | Difficulty: moderate

Q3.

Answer: B

Solution:

Correct Answer: B

Subject: Textile Fibers | Topic: Natural Fibers | Difficulty: moderate

Q4.

Answer: A

Solution:

Correct Answer: A

Subject: Yarn Manufacturing | Topic: Spinning | Difficulty: moderate

Q5.

Answer: A

Solution:

Correct Answer: A

Subject: Fabric Manufacturing | Topic: Weaving | Difficulty: moderate

Q6.

Answer: B

Solution:

Correct Answer: B

Subject: Textile Testing | Topic: Properties | Difficulty: difficult

Q7.

Answer: B

Solution:

Correct Answer: B

Subject: Chemical Processing | Topic: Dyeing | Difficulty: easy

Q8.

Answer: B

Solution:

Correct Answer: B

Subject: Apparel Engineering | Topic: Pattern Making | Difficulty: moderate

Q9.

Answer: C

Solution:

Correct Answer: C

Subject: Textile Machinery | Topic: Machines | Difficulty: moderate

Q10.

Answer: A

Solution:

Correct Answer: A

Subject: Textile Physics | Topic: Friction | Difficulty: easy

Q11.

Answer: B

Solution:

$(1+1/n)^{2n} = [(1+1/n)^n]^2 \rightarrow e^2$ as $n \rightarrow \infty$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q12.

Answer: C

Solution:

Rolle's: f continuous on $[a,b]$, differentiable on (a,b) , $f(a)=f(b)$. Then $f'(c)=0$ for some c in (a,b) .

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q13.

Answer: A

Solution:

Spinning: draws and twists fibers into yarn.

Subject: Textile Manufacture | Topic: Textile Manufacture | Difficulty: moderate

Q14.

Answer: C

Solution:

$\frac{d}{dx}[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1 + 2 = 3$. Hmm. Let me recalc: $\frac{d}{dx} \ln(\sec x) = \tan x$. $\frac{d}{dx} \ln(\tan x) = \sec^2 x / \tan x = 1/(\sin x \cos x) = 2\csc(2x)$. At $\pi/4$: $1 + 2 = 3$. Not in options. The derivative of $\ln(\sec x * \tan x) = \frac{d}{dx}[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1+2=3$. Hmm.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q15.

Answer: A

Solution:

Weaving: interlacing warp (lengthwise) and weft (crosswise) yarns.

Subject: Textile Manufacture | Topic: Textile Manufacture | Difficulty: easy

Q16.

Answer: A

Solution:

Cotton: ~8% moisture regain at standard conditions. Wool: ~16%.

Subject: Textile Fibers | Topic: Textile Fibers | Difficulty: moderate

Q17.

Answer: B

Solution:

$|x|$ is continuous everywhere. Left derivative at 0 = -1, right = +1. Not differentiable at 0.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q18.

Answer: B

Solution:

Cotton: natural cellulosic (vegetable) fiber from plant.

Subject: Textile Fibers | Topic: Textile Fibers | Difficulty: moderate

Q19.

Answer: A

Solution:

$f(x)=x$ is odd on symmetric interval. Fourier series has only sine terms.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q20.

Answer: B

Solution:

Wool: protein (keratin) fiber from sheep. Animal fiber.

Subject: Textile Fibers | Topic: Textile Fibers | Difficulty: difficult

Q21.

Answer: B

Solution:

Nylon: synthetic polyamide fiber. First fully synthetic fiber.

Subject: Textile Fibers | Topic: Textile Fibers | Difficulty: difficult

Q22.

Answer: A

Solution:

$$L\{t^*f(t)\} = -d/ds(F(s)). F(s) = 1/(s+2). -d/ds[1/(s+2)] = 1/(s+2)^2.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q23.

Answer: B

Solution:

$$P(A|B) = P(A \cap B)/P(B) = 0.2/0.4 = 0.5.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q24.

Answer: A

Solution:

$$L\{e^{(-at)}\} = 1/(s+a). \text{ Here } a=3: 1/(s+3).$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q25.

Answer: B

Solution:

$$f(0)=2, f(1)=0, f(-1)=4, f(2)=4. f(-2)=-8+6+2=0. \text{ Roots at } x=1 \text{ (double?) and } x=-2. f(x)=(x-1)^2(x+2). \text{ Two distinct real roots.}$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q26.

Answer: A

Solution:

$$|<u,v>| \leq ||u|| * ||v||. \text{ Equality when } u \text{ and } v \text{ are linearly dependent.}$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q27.

Answer: C

Solution:

Gauss-Seidel converges if matrix is diagonally dominant. Jacobi needs symmetric positive definite.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q28.

Answer: C

Solution:

$$\text{Kirchhoff: for } K_n, \text{ spanning trees} = n^{n-2} = 4^2 = 16.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q29.

Answer: A

Solution:

$$\text{Standard normal: } P(-1 < Z < 1) \text{ approx } 0.68.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q30.

Answer: C

Solution:

$$y = A*\cos(2x) + B*\sin(2x). A=1, B=0. y(\pi/2) = \cos(\pi) = -1.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q31.

Answer: A

Solution:

$$\text{Eigenvalues of } A^2 \text{ are } 1, 4, 9. \text{ Trace} = \text{sum} = 1+4+9 = 14.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q32.

Answer: A

Solution:

$$\text{Area} = \pi*r^2. \text{ Circumference} = 2*\pi*r.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q33.

Answer: A

Solution:

$$P(X \geq 2) = 1 - e^{-4}(1+4) = 1 - 5e^{-4} \approx 0.908.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q34.

Answer: A

Solution:

Row 2 = 2*Row 1, Row 3 = 3*Row 1. All rows linearly dependent. Rank = 1.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q35.

Answer: B

Solution:

$dy/y = dx/x$. $\ln(y) = \ln(x) + C$. $y = C*x$. Wait: $y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow \ln y = \ln x + C \Rightarrow y = C*x$. That's option A. Hmm, let me check: $y \, dx - x \, dy = 0 \Rightarrow y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow y = C*x$. Answer: A.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q36.

Answer: A

Solution:

Eigenvalues are 3 and 1. $[1,0]^T$ is eigenvector for eigenvalue 1. $A^{50}[1,0]^T = [1,0]^T$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q37.

Answer: A

Solution:

PGF of Poisson(λ): $G(z) = E[z^X] = \exp(\lambda(z-1))$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q38.

Answer: A

Solution:

Poisson: mean = λ = variance. Both = 4.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q39.

Answer: A

Solution:

$$f(i) = i^2 + 3i + 2 = -1 + 3i. \operatorname{Re}(f(i)) = -1.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q40.

Answer: A

Solution:

Homogeneous ODE. Substituting $y=x*v$ and $y(1)=0$ gives $y=0$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q41.

Answer: B

Solution:

$$\operatorname{proj} = ((v \cdot w)/(w \cdot w)) * w = (4/5) * (2, 1, 0) = (8/5, 4/5, 0).$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q42.

Answer: A

Solution:

Cotton: ~8% moisture regain at standard conditions. Wool: ~16%.

Subject: Textile Fibers | Topic: Textile Fibers | Difficulty: difficult

Q43.

Answer: A

Solution:

Spinning: draws and twists fibers into yarn.

Subject: Textile Manufacture | Topic: Textile Manufacture | Difficulty: easy

Q44.

Answer: B

Solution:

Wool: protein (keratin) fiber from sheep. Animal fiber.

Subject: Textile Fibers | Topic: Textile Fibers | Difficulty: moderate

Q45.

Answer: A

Solution:

Weaving: interlacing warp (lengthwise) and weft (crosswise) yarns.

Subject: Textile Manufacture | Topic: Textile Manufacture | Difficulty: moderate

Q46.

Answer: A

Solution:

Eigenvalues are 3 and 1. $[1,0]^T$ is eigenvector for eigenvalue 1. $A^{50}[1,0]^T = [1,0]^T$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q47.

Answer: A

Solution:

Spinning: draws and twists fibers into yarn.

Subject: Textile Manufacture | Topic: Textile Manufacture | Difficulty: moderate

Q48.

Answer: A

Solution:

Weaving: interlacing warp (lengthwise) and weft (crosswise) yarns.

Subject: Textile Manufacture | Topic: Textile Manufacture | Difficulty: easy

Q49.

Answer: B

Solution:

$dy/y = dx/x$. $\ln(y) = \ln(x) + C$. $y = C \cdot x$. Wait: $y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow \ln y = \ln x + C \Rightarrow y = C \cdot x$. That's option A. Hmm, let me check: $y \, dx - x \, dy = 0 \Rightarrow y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow y = C \cdot x$. Answer: A.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q50.

Answer: A

Solution:

Cotton: ~8% moisture regain at standard conditions. Wool: ~16%.

Subject: Textile Fibers | Topic: Textile Fibers | Difficulty: easy

Q51.

Answer: B

Solution:

Wool: protein (keratin) fiber from sheep. Animal fiber.

Subject: Textile Fibers | Topic: Textile Fibers | Difficulty: difficult

Q52.

Answer: A

Solution:

Cotton: ~8% moisture regain at standard conditions. Wool: ~16%.

Subject: Textile Fibers | Topic: Textile Fibers | Difficulty: difficult

Q53.

Answer: A

Solution:

Weaving: interlacing warp (lengthwise) and weft (crosswise) yarns.

Subject: Textile Manufacture | Topic: Textile Manufacture | Difficulty: easy

Q54.

Answer: A

Solution:

Poisson: mean = lambda = variance. Both = 4.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q55.

Answer: A

Solution:

Spinning: draws and twists fibers into yarn.

Subject: Textile Manufacture | Topic: Textile Manufacture | Difficulty: easy

Q56.

Answer: A

Solution:

Standard normal: $P(-1 < Z < 1)$ approx 0.68.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q57.

Answer: A

Solution:

Spinning: draws and twists fibers into yarn.

Subject: Textile Manufacture | Topic: Textile Manufacture | Difficulty: difficult

Q58.

Answer: B

Solution:

Cotton: natural cellulosic (vegetable) fiber from plant.

Subject: Textile Fibers | Topic: Textile Fibers | Difficulty: easy

Q59.

Answer: A

Solution:

Weaving: interlacing warp (lengthwise) and weft (crosswise) yarns.

Subject: Textile Manufacture | Topic: Textile Manufacture | Difficulty: moderate

Q60.

Answer: A

Solution:

Eigenvalues of A^2 are 1, 4, 9. Trace = sum = $1+4+9 = 14$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q61.

Answer: A

Solution:

$L\{e^{(-at)}\} = 1/(s+a)$. Here $a=3$: $1/(s+3)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q62.

Answer: A

Solution:

Weaving: interlacing warp (lengthwise) and weft (crosswise) yarns.

Subject: Textile Manufacture | Topic: Textile Manufacture | Difficulty: difficult

Q63.

Answer: A

Solution:

Spinning: draws and twists fibers into yarn.

Subject: Textile Manufacture | Topic: Textile Manufacture | Difficulty: moderate

Q64.

Answer: B

Solution:

Wool: protein (keratin) fiber from sheep. Animal fiber.

Subject: Textile Fibers | Topic: Textile Fibers | Difficulty: easy

Q65.

Answer: B

Solution:

Cotton: natural cellulosic (vegetable) fiber from plant.

Subject: Textile Fibers | Topic: Textile Fibers | Difficulty: moderate

PAPER BLUEPRINT

Subject Distribution:

Engineering Mathematics: 32 questions, 49 marks
Textile Manufacture: 12 questions, 22 marks
Textile Fibers: 13 questions, 21 marks
General Aptitude: 1 questions, 1 marks
Yarn Manufacturing: 1 questions, 1 marks
Fabric Manufacturing: 1 questions, 1 marks
Textile Testing: 1 questions, 1 marks
Chemical Processing: 1 questions, 1 marks
Apparel Engineering: 1 questions, 1 marks
Textile Machinery: 1 questions, 1 marks
Textile Physics: 1 questions, 1 marks

Difficulty Distribution:

Easy: 15 | Moderate: 38 | Difficult: 12

Statistics:

Total Questions: 65
Total Marks: 100

This blueprint is generated from the actual question distribution in this paper.